

## BFMC 2025 : Power Board V2 & Encoder Testing Report

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### Overview of Testing

The goal of the following tests was to evaluate the ability of the power board to perform accurate and reliable voltage regulation. All boards were tested under different input voltages from the battery and under different loads on the motor. The following results showcase the performance of the motor and the difference between the commande being sent and the reading from the encoder.

Testing was also done to calibrate the encoder. This calibration was done by making the car move and computing the encoder displacement over a certain period of time. This displacement was then compared with the measured distance that the car had travelled. This enabled us to find a conversion factor between the angular rotation of the shaft and the linear displacement of the car.

### Power Board V2 Testing Results

#### Graphing of results

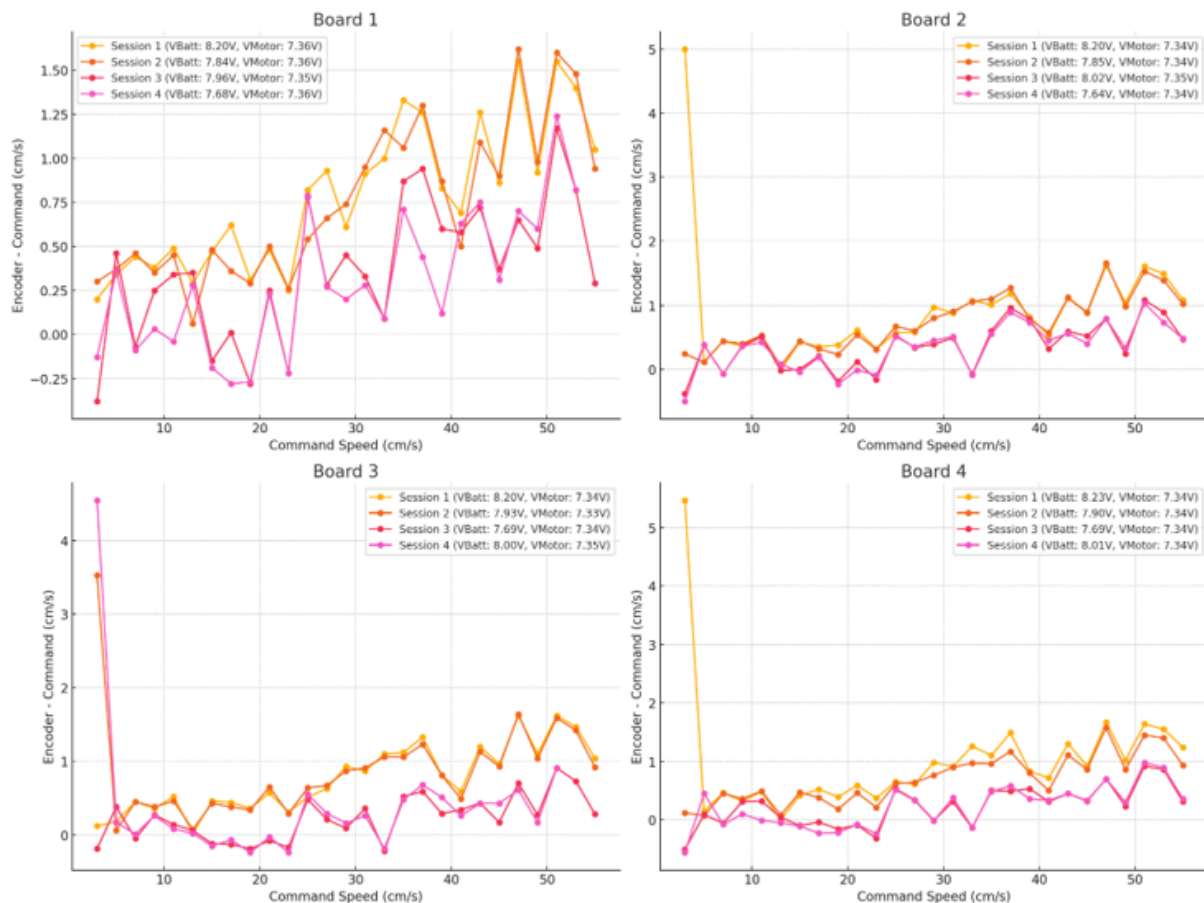


Figure 1: Comparison of input command and encoder reading (Battery = 8.3V)

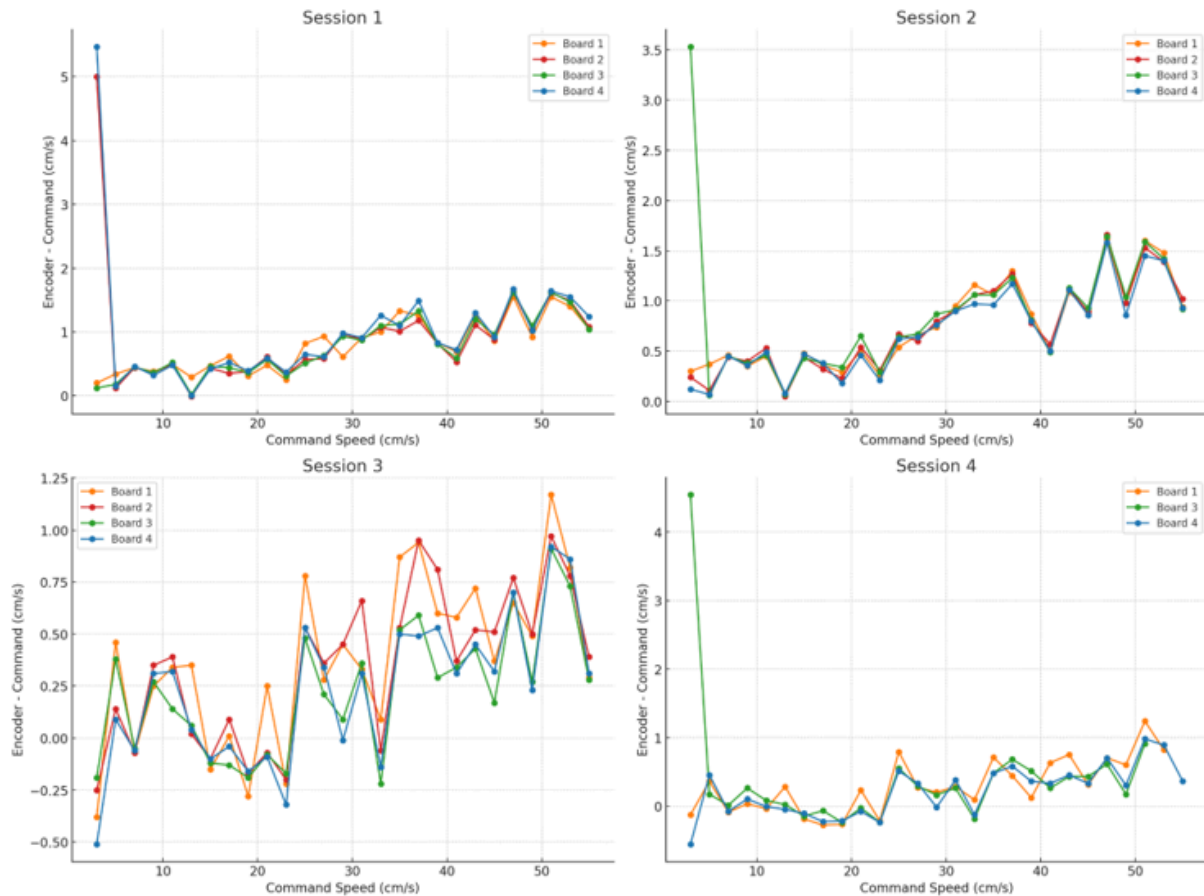


Figure 2: Comparison of input command and encoder reading (Battery = 7.9V)

### Obtained Tables

| Command Speed (cm/s) | Mean Abs Error (cm/s) | Tracking Quality |
|----------------------|-----------------------|------------------|
| 3                    | 1.39                  | ✗ poor           |
| 5                    | 0.25                  | ✓ good           |
| 7                    | 0.26                  | ✓ good           |
| 9                    | 0.3                   | ✓ good           |
| 11                   | 0.36                  | ✓ good           |
| 13                   | 0.09                  | ✓ good           |
| 15                   | 0.28                  | ✓ good           |
| 17                   | 0.28                  | ✓ good           |
| 19                   | 0.27                  | ✓ good           |
| 21                   | 0.33                  | ✓ good           |
| 23                   | 0.25                  | ✓ good           |
| 25                   | 0.61                  | ✗ poor           |
| 27                   | 0.48                  | ✓ good           |
| 29                   | 0.53                  | ✓ borderline     |
| 31                   | 0.63                  | ✗ poor           |
| 33                   | 0.61                  | ✗ poor           |
| 35                   | 0.84                  | ✗ poor           |
| 37                   | 0.99                  | ✗ poor           |
| 39                   | 0.65                  | ✗ poor           |
| 41                   | 0.49                  | ✓ good           |
| 43                   | 0.86                  | ✗ poor           |
| 45                   | 0.63                  | ✗ poor           |
| 47                   | 1.16                  | ✗ poor           |
| 49                   | 0.66                  | ✗ poor           |
| 51                   | 1.3                   | ✗ poor           |
| 53                   | 1.16                  | ✗ poor           |
| 55                   | 0.74                  | ✗ poor           |

Figure 3: Evaluation of the accuracy of input speed command

| Board   | Session | Mean Abs Error (cm/s) | Std Dev (cm/s) | Samples |
|---------|---------|-----------------------|----------------|---------|
| Board 1 | 1       | 0.983                 | 0.349          | 18      |
| Board 1 | 2       | 0.953                 | 0.375          | 18      |
| Board 1 | 3       | 0.550                 | 0.327          | 18      |
| Board 1 | 4       | 0.494                 | 0.345          | 17      |
| Board 2 | 1       | 0.965                 | 0.357          | 18      |
| Board 2 | 2       | 0.955                 | 0.348          | 18      |
| Board 2 | 3       | 0.524                 | 0.323          | 18      |
| Board 2 | 4       | 0.524                 | 0.267          | 18      |
| Board 3 | 1       | 0.988                 | 0.368          | 18      |
| Board 3 | 2       | 0.964                 | 0.349          | 18      |
| Board 3 | 3       | 0.380                 | 0.298          | 18      |
| Board 3 | 4       | 0.387                 | 0.301          | 16      |
| Board 4 | 1       | 1.047                 | 0.376          | 18      |
| Board 4 | 2       | 0.899                 | 0.346          | 18      |
| Board 4 | 3       | 0.409                 | 0.321          | 18      |
| Board 4 | 4       | 0.414                 | 0.318          | 18      |

Figure 4: Statistical metrics of difference between input speed and encoder reading.

## Encoder Testing Results

### Testing with respect to the Power board

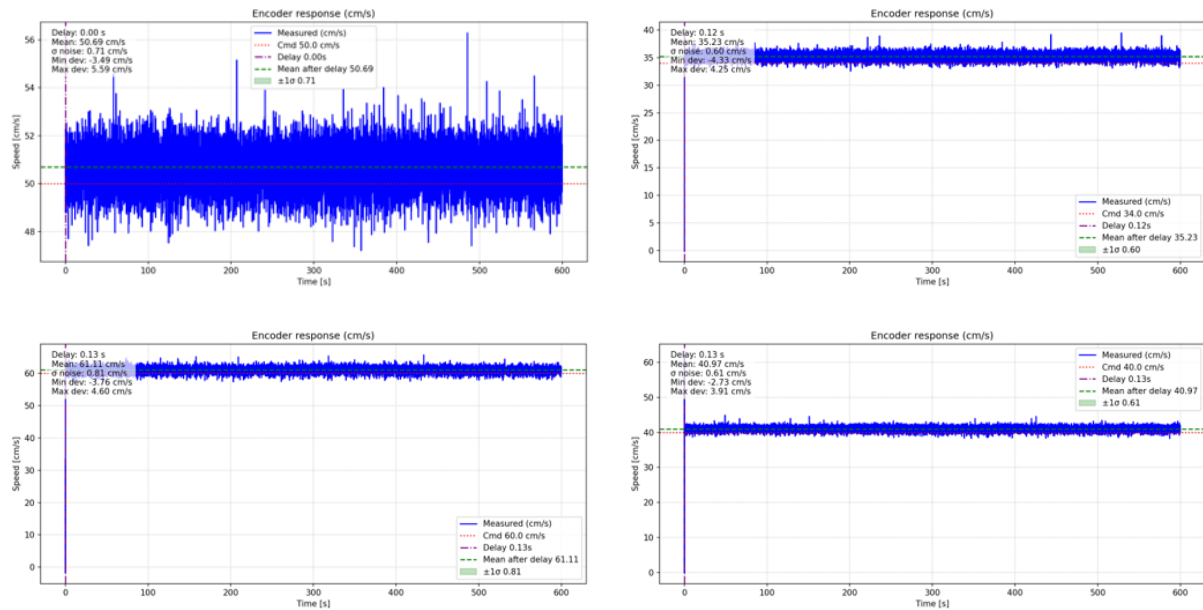


Figure 5: Encoder Noise Readings

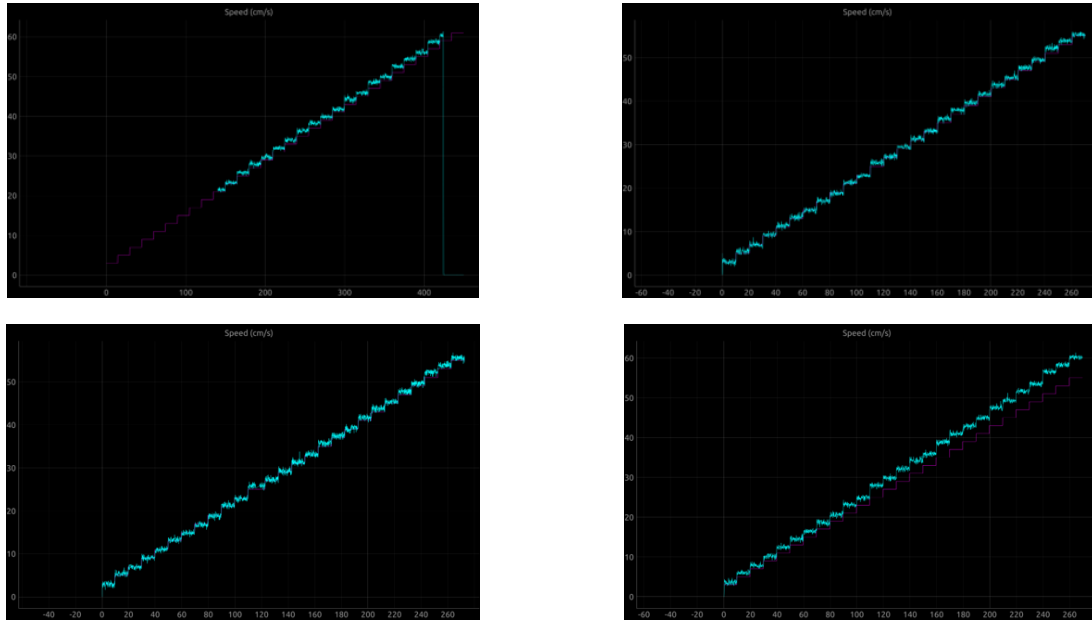


Figure 6: Encoder vs Input Command Tracking

### Encoder Calibration

| Test No. | Speed (mm/s) | Time (s) | Steer Angle (deg) | Encoder Displacement (deg) | Real Measurement (mm) | Distance Traveled (mm) | Factor | Expected Diameter (mm) | Expected Distance (mm) | Expected time |
|----------|--------------|----------|-------------------|----------------------------|-----------------------|------------------------|--------|------------------------|------------------------|---------------|
| 1        | 160          | 20       | 0                 | 45934.7                    | 3165                  | 3165                   | 14.51  | NA                     | 3165                   | 20            |
| 2        | 160          | 30       | 0                 | 69639.63                   | 4835                  | 4835                   | 14.40  | NA                     | 4835                   | 30            |
| 3        | 160          | 26       | 0                 | 59582.71                   | 4120                  | 4120                   | 14.46  | NA                     | 4120                   | 26            |
| 4        | 160          | 25       | 0                 | 58026.98                   | 4010                  | 4010                   | 14.47  | NA                     | 4010                   | 25            |
| 5        | 160          | 25       | 0                 | 57667.89                   | 3970                  | 3970                   | 14.53  | NA                     | 3970                   | 25            |
| 6        | 160          | 18       | 10                | 68033.08                   | 2923                  | 4591                   | 14.82  | 2949                   | 4632                   | 29            |
| 7        | 160          | 16       | 12                | 56715.12                   | 2481                  | 3897                   | 14.55  | 2446                   | 3843                   | 24            |
| 8        | 160          | 13       | 14                | 46980.82                   | 2059                  | 3234                   | 14.52  | 2086                   | 3276                   | 20            |
| 9        | 160          | 11       | 16                | 41793.39                   | 1801                  | 2829                   | 14.77  | 1813                   | 2849                   | 18            |
| 10       | 160          | 10       | 18                | 36641.57                   | 1556                  | 2444                   | 14.99  | 1600                   | 2514                   | 16            |
| 12       | 160          | 14       | 12                | 49439.07                   | 2170                  | 3409                   | 14.50  | 2446                   | 3843                   | 24            |
| 13       | 160          | 21       | -10               | 76238.04                   | 3295                  | 5176                   | 14.73  | -2949                  | -4632                  | -29           |
| 14       | 160          | 9        | -20               | 34161.44                   | 1472                  | 2312                   | 14.77  | -1429                  | -2244                  | -14           |
| 15       | 160          | 9        | 20                | 32817.34                   | 1405                  | 2207                   | 14.87  | 1429                   | 2244                   | 14            |
| 16       | 160          | 15       | -14               | 53433.95                   | 2330                  | 3660                   | 14.60  | -2086                  | -3276                  | -20           |
| 17       | 160          | 11       | -18               | 38198.5                    | 1682                  | 2642                   | 14.46  | -1600                  | -2514                  | -16           |

Figure 7: Table for calibrating the conversion factor for the encoder.

### Interpretation

From the above tests, we are able to see that the motor is able to regulate the voltage accurately. However, we do notice some significant discrepancies between the input speed command and the computation of the encoder. This is because the input speed command was calibrated for a full battery (8.3V) and not the voltage regulated output at the motor (7.34V). This means that a calibration of the motor at an input voltage of 7.34V is required.

For the encoder, Figure 7 shows us that the conversion factor between an angular displacement and the linear displacement is mostly constant no matter if the car is going in a straight line, turning left or turning right. Taking the average of all computed values, we chose a conversion factor of 14.6 degrees per centimeter traveled.